

Copper Weight & Current Capacity

½oz • 1oz • 2oz • 3oz • Heavy copper — and the current they carry

Pick the right copper thickness — or your traces become fuses

Audience: Intermediate Engineering Students

Cu weights • Trace width • IPC-2152 • Temp rise • Heavy copper

Why Copper Weight Matters

It's the biggest variable in current handling

Current capacity

Doubling Cu = $\sim 1.4\times$ current
at same trace width.

Voltage drop

Thicker Cu $\rightarrow$ less IR drop
over long traces.

Thermal handling

Thick Cu spreads heat
better — heatsinks the chip.

Cost

1oz = baseline
2oz = +20-40% cost
Heavy Cu = +200-400%

Etching

Thicker Cu = wider min
trace + spacing required.

Manufacturability

Standard 1oz everywhere.
Heavy Cu = specialty fabs.

Cu Weight — Industry Definitions

Weight per area = thickness

Weight (oz/ft²)	Thickness (µm)	Thickness (mil)	Common use
¼ oz	8.7 µm	0.34 mil	Inner layers in HDI
½ oz	17.5 µm	0.69 mil	Inner layers standard
1 oz	35 µm	1.4 mil	Outer layers standard (default)
2 oz	70 µm	2.8 mil	Power boards, automotive
3 oz	105 µm	4.1 mil	Industrial power, motor drivers
4 oz	140 µm	5.5 mil	Heavy copper begins here
6 oz	210 µm	8.3 mil	Heavy copper standard
10 oz	350 µm	13.8 mil	Bus bars, EV inverters
15-30 oz	525-1050µm	20-40 mil	Industrial DC bus, ultra-heavy

Why is it called 'Weight' not 'Thickness'?

Historic IPC convention — and how to convert

- ▶ IPC standard defines copper layer by oz/ft² — the WEIGHT of Cu on a 1 ft² area
- ▶ Conversion: 1 oz/ft² = 35 μm thick = 1.4 mil thick
- ▶ Most fabs (and KiCad) accept both notation — 1oz / 35μm / 1.4mil all mean the same
- ▶ Outer layers and inner layers can have different Cu weight in the same stackup
- ▶ Common stackup: 1oz outers + ½oz inners (saves cost on multilayer)
- ▶ Heavy Cu typically only applied to outer layers (where current carrying paths live)

IPC-2152 — Trace Width vs Current

Industry-standard current capacity tables

Current capacity (A) at 10 °C temperature rise above ambient — external (outer) layers, 1 oz Cu:

Trace width	Current (10°C rise)	Current (20°C rise)	Current (40°C rise)
6 mil (0.15 mm)	0.5 A	0.7 A	1.0 A
10 mil (0.25 mm)	0.8 A	1.2 A	1.7 A
15 mil (0.38 mm)	1.2 A	1.7 A	2.4 A
20 mil (0.5 mm)	1.5 A	2.2 A	3.1 A
30 mil (0.76 mm)	2.2 A	3.1 A	4.4 A
50 mil (1.27 mm)	3.3 A	4.7 A	6.6 A
100 mil (2.5 mm)	5.6 A	7.9 A	11.0 A
200 mil (5.0 mm)	9.5 A	13.0 A	18.0 A

For 2 oz Cu

Multiply by approximately 1.4× for the same temperature rise. For 3 oz, multiply by 1.7×.

Inner vs Outer Layer Current Capacity

Inner layers carry ~50% less than outer



Inner vs Outer trace current rules

Outer layers

- ▶ In direct contact with air
- ▶ Better heat dissipation
- ▶ Full IPC-2152 chart applies
- ▶ Typical Cu: 1 oz
- ▶ Used for: connector breakouts, power inputs

Inner layers

- ▶ Surrounded by FR-4 (poor heat dissipation)
- ▶ Carry ~50% less current at same width + temperature rise
- ▶ Typical Cu: ½ oz (saves cost)
- ▶ Used for: ground planes, power planes, internal routing
- ▶ For higher current — increase Cu weight or trace width significantly

Voltage Drop Calculations

Long traces + low Cu = wasted volts

- ▶ Resistance per unit length: $R = \rho \times L / (W \times T)$
- ▶ ρ = copper resistivity = $1.68 \times 10^{-8} \Omega \cdot \text{m}$
- ▶ Example: 10 mil trace, 1 oz Cu, 100 mm long, 1 A current
- ▶ $R = (1.68\text{e-}8 \times 0.1) / (0.000254 \times 0.000035) = 0.19 \Omega$
- ▶ $V_{\text{drop}} = 1 \text{ A} \times 0.19 \Omega = 0.19 \text{ V}$ — significant for low-voltage rails!
- ▶ Rule of thumb: keep voltage drop < 1% of nominal voltage
- ▶ For 3.3 V rail: < 33 mV; for 12 V rail: < 120 mV
- ▶ Use 4-wire (Kelvin) sensing for very low voltages (e.g., 0.5 V VCC_CORE)

Trace Width Calculator Tools

Don't compute by hand — use the tools

KiCad built-in

Tools → Calculator Tools
→ Track Width

Saturn PCB Toolkit

Free, comprehensive
More options than KiCad's

Polar Si9000

Industry standard
Licensed

Online calculators

JLCPCB, PCBWay,
Advanced Circuits — free

IPC-2152 software

Official, official
~\$50k license

Hand calc

$\rho \times L / (W \times T)$
+ table for current

Heavy Copper PCBs

When 1-2 oz isn't enough

- ▶ Definition: Cu weight ≥ 4 oz (140 μm) on outer or inner
- ▶ Used in: motor drivers, EV inverters, solar combiners, welding gear
- ▶ Carries: 20-200 A per trace, depending on width
- ▶ Combined with: thermal vias + heatsinks for full thermal solution
- ▶ Cost: 2-5 $\times$ standard PCB (specialty fabrication)
- ▶ Lead time: 3-6 weeks (fewer fabs offer it)

Etching challenge

Heavy Cu requires longer etch times. Result: trace edges become slanted (undercut).
Minimum trace width grows: 1oz = 6mil, 4oz = 15mil, 10oz = 30mil typical.

Indian sources

Heavy Cu specialty fabs in India: AT&S, Centum, Genus.
Most low-cost fabs (JLCPCB, PCBPower) cap at 2-3 oz outer.

Heavy Copper Applications

Where it's mandatory

EV inverters

200-400 A bus connections
6-10 oz typical

Solar combiners

30-100 A DC strings
3-4 oz typical

Motor drivers

10-50 A BLDC drive
2-4 oz

Welding power

100-200 A DC
6+ oz

Telecom power

Battery + rectifier paths
3-4 oz

Audio amplifiers

Class-D output stages
2-3 oz

Mixed Copper Weight Design

Different Cu thickness on different layers

- ▶ Modern fabs support different Cu weights per layer in same stackup
- ▶ Common pattern: 2 oz outer (current-carrying) + ½ oz inner (signal/control)
- ▶ Power layers can be 2-4 oz; signal layers 1 oz or ½ oz
- ▶ Specify in stackup document: 'L1: 2 oz, L2: 0.5 oz, L3: 0.5 oz, L4: 2 oz'
- ▶ Fab compensates for impedance — controlled-impedance traces may need width recalc
- ▶ Cost: 1.3-1.8× standard PCB
- ▶ Lead time: same as standard for most fabs
- ▶ Good for: hybrid power+signal designs (motor controller, charging IC, power IC)

Cu Weight vs Min Trace + Spacing

Thicker Cu = wider rules

Cu weight	Min trace width	Min trace spacing	Min annular ring
½ oz (17 µm)	4 mil (0.10 mm)	4 mil	4 mil
1 oz (35 µm)	6 mil (0.15 mm)	6 mil	5 mil
2 oz (70 µm)	8 mil (0.20 mm)	8 mil	8 mil
3 oz (105 µm)	12 mil (0.30 mm)	12 mil	10 mil
4 oz (140 µm)	15 mil (0.38 mm)	15 mil	15 mil
6 oz (210 µm)	20 mil (0.50 mm)	20 mil	20 mil
10 oz (350 µm)	30 mil (0.76 mm)	30 mil	30 mil

Selection Workflow

Pick Cu weight in 3 questions

- ▶ 1. Max current per trace?
 - ▶ • < 1 A → 1 oz outer is fine
 - ▶ • 1-5 A → 1 oz + careful trace width (≥ 50 mil)
 - ▶ • 5-20 A → 2 oz outer
 - ▶ • 20-100 A → 3-6 oz heavy
 - ▶ • > 100 A → 6+ oz + bus bars / planes
- ▶ 2. Voltage drop budget?
 - ▶ • Low voltage rail (1.2-3.3 V) → tight budget → thicker Cu or shorter traces
 - ▶ • High voltage (12+ V) → looser budget → 1 oz usually fine
- ▶ 3. Layer count and stackup
 - ▶ • 2-layer → outer Cu carries everything → may need 2 oz
 - ▶ • 4-layer+ → inner power planes carry → ½-1 oz inner often enough

Common Copper Weight Mistakes

What lab tests reveal

- ▶ 10 A motor controller on 1 oz Cu with 10 mil trace → trace becomes fuse
- ▶ USB-PD at 5 A on 6 mil 1 oz trace → 1 V drop, port voltage sags
- ▶ Forgetting inner-layer current capacity $\approx$ half of outer-layer rating
- ▶ Using heavy Cu but keeping 6 mil min trace rule → fab can't etch reliably
- ▶ Mixing 2 oz + 1 oz on stackup but using 50 Ω trace width for 1 oz throughout
- ▶ Designing for ambient = 25 °C; product runs at 60 °C → derate by 50%
- ▶ Not specifying outer vs inner Cu weight in fab quote → default to fab choice
- ▶ Heavy Cu but standard fab → fab refuses or accepts then fails QC

Key Takeaways

Key takeaways from this lecture

Cu weight = current capacity

Pick by amperage budget.

1 oz = default

95% of designs work with this.

2 oz for > 5 A

Power boards, motors, USB-PD ≥ 3 A.

Heavy Cu = specialty

Confirm fab capability first.

Inner $\neq$ outer

Inner layer current $\approx$ half.

Use IPC-2152 + tools

Don't guess.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ IPC-2152 — Current-carrying capacity standard
- ▶ Saturn PCB Toolkit — free trace-width calculator
- ▶ KiCad Tools → Calculator → Track Width (built-in)
- ▶ Henkel / Bergquist heavy-copper PCB app notes
- ▶ AT&S India / Centum — heavy-copper specialty Indian fabs
- ▶ JLCPCB / PCBWay — confirm max Cu weight supported
- ▶ Phil's Lab — heavy copper PCB video walkthroughs
- ▶ Robert Feranec — power-stage layout videos

Thank you • Build something this week.